The empirical recurrence for OEIS A299726, recovered from its tail

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Abstract

OEIS A299726 records "Empirical recurrence of order 92" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 108$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 94$. Its last coefficient is $c_{92} = -165888000 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299726 is "Number of nX6 0..1 arrays with every element equal to 1, 2, 3, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$5, 888, 57471, 3224826, 187530622, 10997877289, 643936117646, 37689687396652, \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above)

As of the "Last modified" line on the live entry (Feb 17 2018 10:55:06, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 108$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 108$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 216$ exact terms from $n = 2$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_1	43	c_{24}	−2959708627372067	c_{47}	−1179270768168441115 23	c_{70}	51635528816301
c_2	642	c_{25}	17064698374637647	c_{48}	−3641015607327482853 66	c_{71}	6458817493027
c_3	12604	c_{26}	9339158731707501	c_{49}	−2788908381324899552 23	c_{72}	2791006670701
c_4	159467	c_{27}	38980672031659370	c_{50}	−531245363746507138 27	c_{73}	−279192570775
c_5	1026589	c_{28}	228772805955243117	c_{51}	1604249561741205781 84	c_{74}	−295371804321
c_6	4055868	c_{29}	47609088780189858	c_{52}	3178166654461027600 80	c_{75}	−367350572856
c_7	−15466823	c_{30}	−54180816528635239	c_{53}	2156247099955311993 68	c_{76}	−848114792801
c_8	−382074477	c_{31}	−630376035737091074	c_{54}	346454163172850321 54	c_{77}	−74122732456
c_9	−1917659749	c_{32}	−3433186619252539667	c_{55}	−1201912983596333081 35	c_{78}	286206980734
c_{10}	−585867988	c_{33}	−2241329413721940937	c_{56}	−1685719712056037192 45	c_{79}	115260398714
c_{11}	37909067780	c_{34}	162474199242981948	c_{57}	−1014406088420171944 37	c_{80}	412605467371
c_{12}	230874659015	c_{35}	4872480855928054361	c_{58}	−201326531715716979 77	c_{81}	−34661925094
c_{13}	283905527807	c_{36}	25487389112057255190	c_{59}	458788444244076806 61	c_{82}	96572589454
c_{14}	−3338396240071	c_{37}	20723817873639157401	c_{60}	532783882808491385 69	c_{83}	−25949044830
c_{15}	−12596102643117	c_{38}	2221760255307501932	c_{61}	324254580077175856 20	c_{84}	−3470449387
c_{16}	5734894160961	c_{39}	−20256890648880227011	c_{62}	109471625650247440 27	c_{85}	87660503747
c_{17}	84446162679807	c_{40}	−104073606569449823314	c_{63}	−806334501487610072 6	c_{86}	−3225485408
c_{18}	130189039966994	c_{41}	−89882378432385571998	c_{64}	−971043098518194142 1	c_{87}	43731034188
c_{19}	117201975558841	c_{42}	−15857099169461917419	c_{65}	−718222704964537952 6	c_{88}	12459361152
c_{20}	−399879953782427	c_{43}	57915179644426947902	c_{66}	−354047641583818389 5	c_{89}	−1646813542
c_{21}	−2016694263908176	c_{44}	251594590323158598535	c_{67}	317495910194813522	c_{90}	−773360640
c_{22}	−1289533224086516	c_{45}	212049848317229896484	c_{68}	765264937353802651	c_{91}	−871833600
c_{23}	−970090804779726	c_{46}	44580335536382048011	c_{69}	866649457367416528	c_{92}	−165888000

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 94$, and it is the recurrence OEIS A299726 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1 x^{92-1} - \cdots - c_{92}$. Its constant term is $-c_{92} = 165888000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -165888000.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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